



Time: 5 minutes

Outcome: Use marginal productivity to analyze labor demand, wages, and employment.

Difficulty: Intermediate

Factor Markets & Labor Demand

THE CORE IDEA

Labor demand is derived from the value workers add to output. Marginal product of labor is the extra output from one more worker. For a price-taking seller, $VMP = \text{output price} \times MPL$. A profit-maximizing firm hires labor up to the point where VMP equals the wage.

HOW TO RECOGNIZE IT

- A wage change causes movement along labor demand; productivity or output-price changes shift labor demand.
- Higher product demand, output price, or worker productivity shifts labor demand right.
- Worker preferences, population, migration, training, and alternative jobs can shift labor supply.
- Competitive wage and employment occur where market labor demand intersects market labor supply.

WATCH OUT

Do not confuse MPL with VMP. MPL is extra physical output; VMP converts that output into dollars by multiplying by the output price.

WORKED EXAMPLE: COMPETITIVE LABOR-MARKET EQUILIBRIUM



Suppose output sells for \$5 and the marginal worker adds 4 units per hour. VMP is $\$5 \times 4 = \20 per hour, matching a \$20 wage. The graph shows market equilibrium at that wage and 40 hundreds of workers, or 4,000 workers. A higher output price raises VMP at each employment level and shifts labor demand right.

CHECK YOURSELF

If the output price rises while worker productivity is unchanged, what happens to labor demand?